

AVINASH PANDEY

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EDUCATION

Northeastern University

Expected May 2028

Master of Science in Artificial Intelligence, Incoming Fall 2026

Planned Coursework: Foundations of Artificial Intelligence, Linear Algebra and Probability for Data Science

Purdue University

December 2024

Bachelor of Science in Computer Science, GPA: 3.3/4.0

Relevant coursework: Artificial Intelligence, Data Science, Database Systems, Operating Systems, Theory of Computation

TECHNICAL SKILLS

Languages: Python, SQL, Java, R

AI/ML: PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, NumPy, Pandas

LLM & RAG Systems: LangChain, OpenAI API, Gemini API, Claude API, FAISS, ChromaDB

Data & MLOps: PySpark, Docker, Kubernetes, AWS EC2, AWS S3, AWS Lambda, MLflow, Git/GitHub

Development Tools & Visualization: Streamlit, Tableau, Power BI, Cursor, Claude Code, GitHub Copilot

PUBLICATIONS

Manuscript Preprint

August 2025

Pandey, A.; Kuznetsov, A.; Mukhopadhyay, S.

- Multi-Model LLM Architectures for Personalized Summarization and Relevance Ranking in Biomedical Literature.*
First-author manuscript available as a bioRxiv preprint.

RESEARCH EXPERIENCE

Purdue University

Feb 2025 – Feb 2026

Research Assistant – Advisors: Dr. Snehasis Mukhopadhyay, Dr. Alexey Kuznetsov

Indianapolis, IN

- Designed a multi-LLM summarization pipeline using Google Gemini 2.0 Flash and GPT-4o-mini to generate personalized biomedical literature summaries, achieving BERTScore-F1 of 0.86 across 20 biomedical queries.
- Built an asynchronous retrieval and metadata-enrichment workflow using PubMed Entrez and NIH iCite to collect article metadata, citation signals, and relevance features across biomedical literature.
- Developed a citation-aware ranking approach combining ontology-guided keyword extraction, cosine similarity, and Relative Citation Ratio (RCR) to surface high-value papers.
- Ran ranking and retrieval ablations comparing TF-IDF, BM25, ontology-constrained keywording, and hybrid citation-similarity scoring to analyze trade-offs in relevance quality.
- Implemented multi-model aggregation and validation steps to compare generated summaries, reduce unsupported claims, and improve reliability for user-facing literature review workflows.
- Built a Streamlit-based research interface for query input, candidate paper review, ranked result inspection, and summary generation during user evaluation.
- Evaluated the system with ROUGE, BERTScore, ablation studies, and a 10-user study; documented methods and findings in a first-author biomedical AI preprint.

PROFESSIONAL EXPERIENCE

Legislative Services Agency

Dec 2023 – Mar 2024

Business Analyst Intern

Indianapolis, IN

- Designed and managed ETL workflows to validate and preprocess Indiana General Assembly datasets, achieving 98% accuracy across legislative bill and report datasets.
- Developed and optimized SQL-based workflows for extraction, transformation, validation, and reporting within SQL Server.
- Created Tableau and Power BI reporting views that helped analysts inspect legislative data quality, summarize bill activity, and communicate trends to non-technical stakeholders.
- Collaborated with Business Analysts and Software Developers to troubleshoot data integration issues and maintain reliable access to structured legislative data sources.

PROJECTS

ICML 2024 Research Paper Analyzer <i>Personal Project, LLM/RAG Pipeline</i>	Jul 2024 – Nov 2024
• Built an end-to-end AI pipeline to collect, parse, retrieve, summarize, and analyze 2,600+ ICML 2024 research papers using PDF extraction, vector search, and LLM-based summarization. • Implemented retrieval workflows with FAISS and ChromaDB, enabling paper-level question answering and topic exploration over a large research corpus. • Used Gemini 1.5 Flash and LangChain metadata filtering to classify research areas, extract structured paper insights, and support targeted analysis by topic, author, and institution. • Developed dashboard visualizations to surface research trends, top contributors, institutional patterns, and major topic areas for non-technical users exploring AI research and startup opportunities. • Documented pipeline assumptions, search examples, and limitations to support reproducibility and handoff of the research-paper analysis workflow.	
Mutual Learning Algorithm for News Classification <i>Capstone Project, NLP, ML</i>	Aug 2024 – Dec 2024
• Implemented a mutual-learning setup where heterogeneous models, including Naive Bayes, SVM, and MLP, iteratively co-taught on labeled splits and pseudo-labeled unlabeled news data. • Built an NLP preprocessing and evaluation pipeline with tokenization, stop-word removal, stemming/lemmatization, accuracy, precision, recall, and F1 analysis. • Compared teacher-student pairings to understand when semi-supervised learning improves classification performance and where pseudo-labeling introduces failure cases. • Documented experimental setup, evaluation results, and model behavior to support reproducible follow-up experiments on semi-supervised text classification.	
AI-Driven Diabetes Prediction Pipeline <i>Python, scikit-learn, Deep Learning</i>	Aug 2024 – Nov 2024
• Built a supervised ML pipeline on a public Kaggle diabetes dataset, including exploratory analysis, outlier handling, categorical encoding, and class rebalancing with SMOTE/SMOTE-Tomek. • Benchmarked Decision Tree, Random Forest, SVM, and MLP models with cross-validated hyperparameter search, emphasizing recall and ROC-AUC for the screening context. • Used feature diagnostics to highlight HbA1c and blood glucose as primary signals and documented trade-offs between classical models and neural baselines.	
Counts of Bushels Project <i>SQL, SQL Server 2022, Data Analytics</i>	Aug 2023 – Oct 2023
• Designed a normalized relational schema for corn and soybean yield data, loaded representative records, and validated integrity with targeted analytical SQL queries. • Created a reusable query suite for common agronomic analyses across yields, regions, and time windows while documenting assumptions and edge cases.	

PRESENTATIONS

Personalized Multi-LLM Summarization for Biomedical Literature	Jul 2025
<i>Purdue University research group talk</i>	<i>Indianapolis, IN</i>
ICML 2024 Research Paper Analyzer: Pipeline and Dashboard Demo	Nov 2024
<i>Project webinar and live demo</i>	<i>Indianapolis, IN</i>

HONORS & AWARDS

- Dean’s List, 5 terms, 2020–2024
- Alpha Lambda Delta & Phi Eta Sigma Honor Society, 2021
- International Jaguars Excellence Award, 2020

LEADERSHIP & SERVICE

Vice President, Computer Science Club	2023–2024
<i>Purdue University</i>	<i>Indianapolis, IN</i>
• Organized hackathons, project nights, and speaker events; increased member engagement by 80% through recurring student-led programming.	
International Peer Mentoring Program, Mentor	2022–2024
<i>Purdue University</i>	<i>Indianapolis, IN</i>
• Mentored incoming international students on academic resources, campus logistics, and cultural adjustment through regular check-ins and referrals.	